



Dual-core Processors

What are dual-core processors?

A dual core processor is two processor cores on one die, essentially like having a dual processor system in one processor.

How will shifting to dual-core help?

The problem with winding up clock speeds is heat. High heat means errors. The heat comes from power. It takes a lot of juice to crank up a processor to high clock speeds and a processor with that much electricity running around the die is prone to electrical noise, akin to interference. The pathways on a processor are very close together. More power running through these pathways means there will be electrical radiation from one pathway to the next. This leakage could corrupt the data, and corrupted data means errors.

Where can I find a dual-core processor?

Expect to see dual-core desktops in India soon!

Why do we need dual-core processors?

A properly-configured dual-core system will react quicker when lots of applications are running simultaneously. The OS can manage which applications are sent to which processor core—so if one core is encoding a TV show, the OS can send your command to open up a Web browser to the second CPU core. In short, a dual-core processor will make your PC multi-task better.

When was the first dual-core processor launched?

The first dual-core processor was IBM's Power4 chip in 2001, beating Intel's Pentium Processor Extreme Edition 840 by a few days.

Who manufactures dual-core processors?

Most major chip manufacturers, including AMD and Intel, are in the dual-core race.

The company already has the SkypeOut feature for making calls from computers to regular phones, which costs less than 2 cents (Rs 0.87) a minute. SkypeIn allows incoming calls. With SkypeIn, users get a regular phone number and can receive calls from a landline or mobile phones without having to pay roaming charges.

Skype Voicemail lets users receive a voice-mail message up to 10 minutes long from any user or traditional phone.

According to Niklas Zennstrom, CEO and co-founder, both packages would give family members and friends of Skype users, who are not connected to the Internet, a way to interact with them.

The number of downloads for the free software has crossed 100 million, with more than two million people using it at any given moment.

Skype, based in Luxembourg and founded in Sweden, said SkypeIn will cost \$13 (Rs 565) for three months and \$40 (Rs 1,740) for a year. A Skype Voicemail subscription costs \$7 (Rs 305) for 3 months and \$20 (Rs 870) for the year.

In related news, unconfirmed reports claimed that Yahoo Inc. was planning on a merger or a complete takeover of Skype.

FLASH ME AGAIN!

Macromedia Unveils Flash Platform

Macromedia unveiled the Macromedia Flash Platform for delivering effective experience for rich content, applications, and communications across browsers, OSes and devices of all kinds.

The Flash Platform is a complete system, including

a universal client runtime, an openly-published file format (.SWF) specification, a programming model, developer tools, integrated solutions, and the support of major systems integration partners, ISVs, and OEMs. A host of third-party tools and solutions based on the Flash platform leverage the platform, such as SAP NetWeaver Visual Composer. The Flash platform enables differentiated experiences in a variety of vertical markets.

The company also previewed the next generation of Macromedia Flash Player, now in beta. Code-named Maelstrom, the updated player will dramatically improve Flash expressiveness and performance. It will further strengthen the Flash Platform with improved font display and a new, higher-quality video codec. Maelstrom will be available for public download later this year.

TICKET TO SAFETY

Microchip In The Passport

The German government is using highly secure chips developed by Infineon Technologies in an effort to make counterfeiting and unauthorised use virtually impossible in future electronic passports, which the German Passport Office intends to issue starting November 2005.

Infineon will also be supplying a special chip package developed for identity cards and passports as well as the inlay containing the antenna and its connection to the chip.



In the new electronic passports, printed information about the holder's identity—such as name, date of birth and photo, and the validity period and number—will also be stored in encrypted format on a chip.

Moreover, from March 2007, fingerprints of the bearer's index fingers will also be stored in an encrypted form. Plans to invisibly integrate the chip into the front cover are also being discussed.

The data stored on the chip can be accessed only when the passport is opened. This data is then transmitted to an authorised and certificated read-write device.

Over 50 individual security mechanisms integrated into the chip technology will help ensure that personal data is protected against unauthorised read-out and manipulation.

The German company Bundesdruckerei will be producing the passports and providing the necessary infrastructure including the background system and even the reading devices. Infineon Technologies is one of the two semiconductor suppliers for the German passport.

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